

Integration Paper

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Looking back at my learning theories paper earlier this semester, I felt that social cognitive theory best reflected my personal approach to teaching. As described by Albert Bandura, social cognitive theory describes three significant factors that influence learning, namely a triad of personal, behavioral, and environmental factors. This trio is known as reciprocal determination or triadic reciprocity causality. At the time of my initial paper, I was initially anticipating creating a course for either medical trainees who had completed medical school, or current Gastroenterology fellows who were rotating with the inpatient Hepatology hospital team. As such, my focuses initially were based on clinical education, bedside examination, and endoscopic skills. Given the importance of observational learning and modeling, Bandura's social cognitive theory was an effective strategy to incorporate into my original design. This theory would be highly applicable to learning as a part of an inpatient medical team or gaining endoscopic skills. For example, enactive learning (Schunk, 2020) would occur by longitudinally following patients during their hospital admission and observing their clinical course, depending on the medical interventions conducted. Observational learning could also occur by demonstrating bedside examination techniques or positive physical exam findings seen in patients followed by the team. Modeling would also occur through endoscopic observation and the practical application of skills gained.

As I continued to develop my curriculum outline, my plans for my course evolved. Instead of focusing on a single rotation for medical trainees, I wanted to pivot and create a 15-week curriculum to be integrated into medical school education. As such, my plans for course content, means of assessment, and locale of instruction changed. I realized that with bedside learning and endoscopy skill education, it would be challenging to integrate online learning mechanisms, which would be better suited for a course directed at medical students. I, therefore, changed directions and designed a curriculum covering a wide range of critical concepts in hepatology, incorporating weekly readings, multiple choice questions, case-based learning, and a final assignment integrating electronic medical records mimicking a realistic clinical scenario. Reviewing the teaching strategy selected for my curriculum, I also incorporated elements of social cognitive theory and largely kept a similar pedagogic ethos to that described in my original learning theories paper, where I also went into depth regarding Bandura's social cognitive theories and triadic reciprocal causation.

One form of assessment during my course would be through weekly open-book multiple-choice questions. This is one scenario that is less effectively conveyed through social cognitive theory and more accurately reflects a constructivist model. Especially with students having the open-book opportunity, I feel this is best described through the zone of proximal development as described by Vygotsky (Schunk, 2020). Though building blocks and core concepts to answer questions would be introduced through the weekly readings, an open-book policy would allow students to work together or use the textbook (or various internet resources) to act as a scaffold to answer questions. Vygotsky's sociocultural theory of language would also aid in learning if students opted to work together to answer these multiple-choice

questions. Though this means of assessment does not quite fit my learning theory, I felt it necessary to include it as this mechanism is often used in medical education (as well as in certification examinations). One could argue that real-life examinations would be better emulated by controlling the learning environment by removing the open-book policy and integrating technology such as a locked browser to answer multiple-choice questions. This change would also better fit Bandura's triadic reciprocal causation theory (especially through the behavioral aspect of emulating a more realistic testing environment). However, I would also like to be mindful of the potential stressors on students by creating a more rigid testing environment – especially if this is a weekly assignment. In re-evaluating my curriculum outline, I would maintain the open-book consideration even though this does not accurately represent social cognitive theory.

Another means of assessment proposed in my curriculum outline is through six problem-solving clinical cases ("Clinical Case Discussion Posts") split throughout the course based on realistic clinical scenarios. Potential clinical examples would include acute liver failure, acute alcohol-associated hepatitis, or liver disease in a pregnant woman. My intention was to provide a clinical case online along with a set of questions related to diagnosis, investigations, and medical interventions. Students would be expected to create discussion posts detailing their responses and have peers weigh in with their thoughts regarding diagnosis or potential management changes. My role as instructor would be to also be active in the discussion board and post replies to help enhance the clinical scenario (such as suggesting changes in clinical status based on a learner's suggested medical interventions) and augment the online learning experience. This assessment fits within social cognitive theory in several regards. Online

discussion boards would facilitate observational learning, and through active instructor involvement, the aim is to improve motivation in learners concerning their approach to the provided clinical scenario. Cognitivism is also demonstrated in this type of assessment through the self-directed learning required to correctly respond to the clinical scenarios and questions posed. In order to better align this assessment with my ideal learning theory, incorporating weekly instructor lectures based on the assigned readings would allow for better integration of social cognitive theory through demonstration of modeling and decreasing a learner's need for additional self-directed learning. Self-efficacy and modeled experiences are essential strategies for learners to acquire knowledge (Ryan, 2012) and fit within social cognitive theory. In retrospect, to better align my assessment with Bandura's theory, I could also consider incorporating videos of volunteers playing out the assigned clinical scenario (both as the medical trainee and as the patient). This would establish further modeling elements and aid in gauging self-efficacy as another component of social cognitive theory (Schunk, 2020). Self-efficacy allows individuals to reflect on their abilities and compare them against the modeled activities demonstrated in videos. This would allow for improved skill development and aid learners in preparing for their final assignment (described below).

My curriculum's final means of assessment is through a comprehensive clinical scenario allowing students to work with a sandboxed electronic medical records system. A clinical scenario would be provided to students, with sample patient records available in an electronic medical records (EMR) system (such as Epic). This final assessment is titled "Clinical Plan from EMR" in my curriculum outline. Learners would be expected to virtually present the patient (using a style commonly used during hospital rounds) and create an individual care plan for this

patient. A sample video would be provided (potentially using a different clinical scenario) with a former course graduate or medical resident conducting a sample patient presentation to allow students to model and compare against. As discussed in my curriculum outline, this would better set expectations for students (incorporating the cognitive factor of Bandura's theory), and the environmental factors would be controlled through the EMR as above. Students would post their presentation as an online video, and peers would provide anonymous feedback. This assessment would incorporate elements of mastery experiences (Ryan, 2012), and anonymous peer feedback would be a source of social persuasion and positive feedback or encouragement. Comparing the described elements of this assessment against the learning theory discussed in my original paper, I feel that this activity encompasses several elements of social cognitive theory and would not make any changes accordingly.

Upon reviewing my learning theories paper, I discussed integrating technology through various means, including Zoom or Microsoft Teams lectures and integrating large-language model (LLM) derived artificial intelligent chatbots such as ChatGPT or Google Bard. Though applicable to social cognitive theory, these elements were ultimately not integrated into my curriculum outline. In retrospect, one addition I could have made is to create online presentations (using audio/visual means) to supplement the weekly assigned readings as discussed above. Not only would this allow me to augment the readings and highlight topics of particular importance, but it would also be a way to demonstrate modeling (a fundamental tenet of social learning theory). Though some elements of modeling are included in my curriculum, such as in the final assignment having former course graduates or medical residents create a sample video or through anonymous peer review, weekly presentations conducted by

the instructor would only serve to reinforce course content and provide another means to aid learner motivation.

Another aspect described in my learning theories paper that was not incorporated in my curriculum outline is that of large language model (LLM)-derived chatbots. Though the technology already exists in an early form and has already undergone several iterative updates over the past several months, the technology does not currently allow for automated chatbot integration into discussion boards. In reflection, I presented an idealized futuristic application of this technology that may become feasible in the months to years to come. One month ago, Microsoft announced that its LLM (formerly known as Bing Chat) would be rebranded as Copilot and will be integrated into Microsoft 365 and Office products. The integration of LLM into ubiquitous software already utilized by students will need to be considered if my curriculum is ultimately implemented to ensure that students generate thoughts and assignments themselves (without artificial intelligence assistance) to ensure that students genuinely learn through the intended learning theories as designed in the course.

Five learning outcomes were included in my course curriculum, encompassing elements from throughout the course. My course outline included a column highlighting the learning outcomes targeted with each week's readings and assessments. The first learning outcome uses a somewhat vague taxonomy verb (that I would likely change with further syllabus revisions) but incorporates hepatic anatomy and physiology while applying this knowledge to various disease processes involving the liver. This outcome is addressed each week from week one through week thirteen of the course, and I feel that utilizing multiple-choice questions is an effective strategy to test students' knowledge of these concepts. As discussed above, these multiple-

choice questions can be completed open book to allow students to utilize the referenced materials or work in groups to complete them. This therefore aligns well with the learning outcome, and I feel would be an effective strategy to gauge understanding.

The second learning outcome discusses applying a student's knowledge of liver disease (acquired through course content and assigned readings) towards provided clinical scenarios and generative therapeutic plans. Six clinical cases are incorporated into the course outline, separate from the final assignment (which serves as a more comprehensive clinical plan along with a video presentation). By utilizing real-life scenarios, based on my clinical experience, these assessments would accurately represent what students can expect to encounter as they begin their clinical rotations and start seeing patients. Through discussion posts and sample cases in a sandboxed EMR, these would be effective strategies for assessing learners' knowledge and satisfying the second learning outcome.

The third learning outcome discusses medical indications for liver transplantation which is integrated in the readings spread out through the course and covered in depth in week twelve. Using multiple-choice questions and the clinical case assignments described earlier, these would be tailored to this learning objective, especially in week twelve of the course (where the learning topic is listed explicitly as "Liver transplant indications and pre-transplant considerations." Weeks thirteen through fifteen are also tailored towards liver transplantation and would reinforce the concepts discussed during week twelve.

My fourth learning outcome pertains to the steps undertaken during transplant surgery and reinforces concepts acquired at the start of the course related to hepatic anatomy and

pathophysiology. This is addressed during weeks thirteen and fifteen and covered through the clinical case assessment and multiple-choice questions assigned at that time.

The final learning outcome is directed towards weeks fourteen and fifteen with the topics of pharmacology of immunosuppression, immunology, and late complications of liver transplantation highlighted as the key topics during that two-week interval. Though the intention of this learning outcome was to be incorporated into several provided clinical cases, there is only one opportunity for such assessment – that being the final comprehensive assessment. The multiple-choice questions can satisfy this learning outcome in part; however, upon retrospective review of my curriculum outline, I would either change this final learning outcome to be a bit more general (i.e., avoid specifying clinical cases in plural) or add an additional clinical case discussion post assignment to be directed towards week fourteen, in particular, to satisfy this learning outcome better.

In summary, the learning theory I selected for my original five-page paper was social cognitive theory, first described by Albert Bandura in the 1960s. As this course progressed, my ideas for my curriculum evolved to better integrate technology for online learning and ultimately created a comprehensive fifteen-week curriculum outline to be integrated into medical school education. Though several components of my curriculum integrate social cognitive theory, there are various potential changes that I have proposed above to modify my curriculum, assessments, and learning outcomes in order to better align with this learning theory. Five learning outcomes were established for this curriculum, and though they are primarily well covered through my varied assessment strategies, my final learning outcome

could be modified (or the curriculum adjusted slightly) to be a more accurate reflection of student expectations.

References:

Ryan, R. M. (2014). *The Oxford Handbook of Human Motivation*. Oxford University Press.

Schunk, D. H. (2020). *Learning theories: An educational perspective (8th ed.)*. Pearson.